

Using LLMs to Reduce Workload for Volunteers Reviewing IHHC Diplomate Submissions

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9/17/2026

Disclosures

- No financial relationships to disclose

Unpacking the Title

- Using LLMs To **Reduce Workload** for Volunteers Reviewing IHHC Diplomate Submissions
 - IHHC Diplomate Submissions
 - As part of CC cycle, diplomates must complete one IHHC/QI project that follows the PDSA methodology every 5 years
 - Reduce Workload
 - 16k diplomates with ~10k participating in CC : nearly 2,000 individual submissions each year

Unpacking the Title

- Using LLMs To **Reduce Workload** for Volunteers Reviewing IHHC Diplomate Submissions
 - Reduce Workload
 - Keep human in the loop, not looking for a fully automated, hands-off process
 - Every submissions is read, reviewed, and evaluated by a human
 - LLMs : Large Language Models that process and generate text
 - ChatGPT, Gemini, Claude, etc.

Opportunity

- Prior to 2026, ABPMR Board Members rotated submission evaluation duty throughout the year
 - Certain periods were busier than others and, at times, the workload was unreasonably high
 - ABPMR Board identified the need to reduce this burden and already planned to make changes for 2026
- ABPMR staff conducted a brief POC for LLM involvement late 2025

Proof-of-Capabilities

- First step was evaluating capabilities of LLMs
 - Gathered Historic Submissions and Determinations
 - Played with some basic LLM guidance and asked for Acceptance or Rejection of submissions
 - Compared results to human determinations
- Done manually with Microsoft 365 CoPilot
 - No software written yet
 - No workflow integration attempted

Proof-of-Concept

- Once capabilities were proven, workflow integration needed to be proven
- Explored integration options with the existing workflow
 - Knew a human would be in the loop
 - Knew goal was reducing burden
- Explored Microsoft 365 CoPilot Integration
- Explored in-house software harness
- Settled on 3rd Party Integration with Submission Platform

Rejected Ideas

- Microsoft 365 CoPilot Integration
 - CoPilot studio had automation/bot creation tools
 - Would require manual triggers
 - Determined it increased workload of staff needlessly
- In-house Software Harness
 - Possible, but would have been first-of-its-kind
 - Determined 3rd Party Solution was faster to implement

Submission Workflow



Submittable

Submittable



American Board of Physical
Medicine and Rehabilitation

IHHC (QI) SUBMISSION PLATFORM

Exercise Is Medicine

Guidelines ▾

Submit Project

Cerebral Palsy Hip Surveillance Program

Guidelines ▾

Submit Project

Publications IHHC

Guidelines ▾

Submit Project

IHHC Group Project

Guidelines ▾

Submit Project

SECTION 1: PLAN

Identify an area in your practice that needs improvement.

A) What is the problem you are trying to solve? *(required)*

What do you want to improve within your practice? Look for inefficiencies, annoyances, or safety issues. Consider complex issues, but focus on simple solutions.

...

SECTION 2: DO

Describe the desired outcomes and the requirements needed to achieve them.

A) What change(s) did you implement? *(required)*

You can implement one change, or you can choose to do several at a time. Be specific about the changes you made in order to solve the problem you identified.

Submission Workflow



Activity
Review
Messages

Type a note ...

6
Assigned Team
Add

all
notes
assignments
ratings
labels

VIEWABLE BY SUBMITTER
Thu, Sep 3, 2026 9:32 AM
David Pruitt sent a message

VIEWABLE BY SUBMITTER
Thu, Sep 3, 2026 9:27 AM
Moved to **Modifications Needed** by David Pruitt

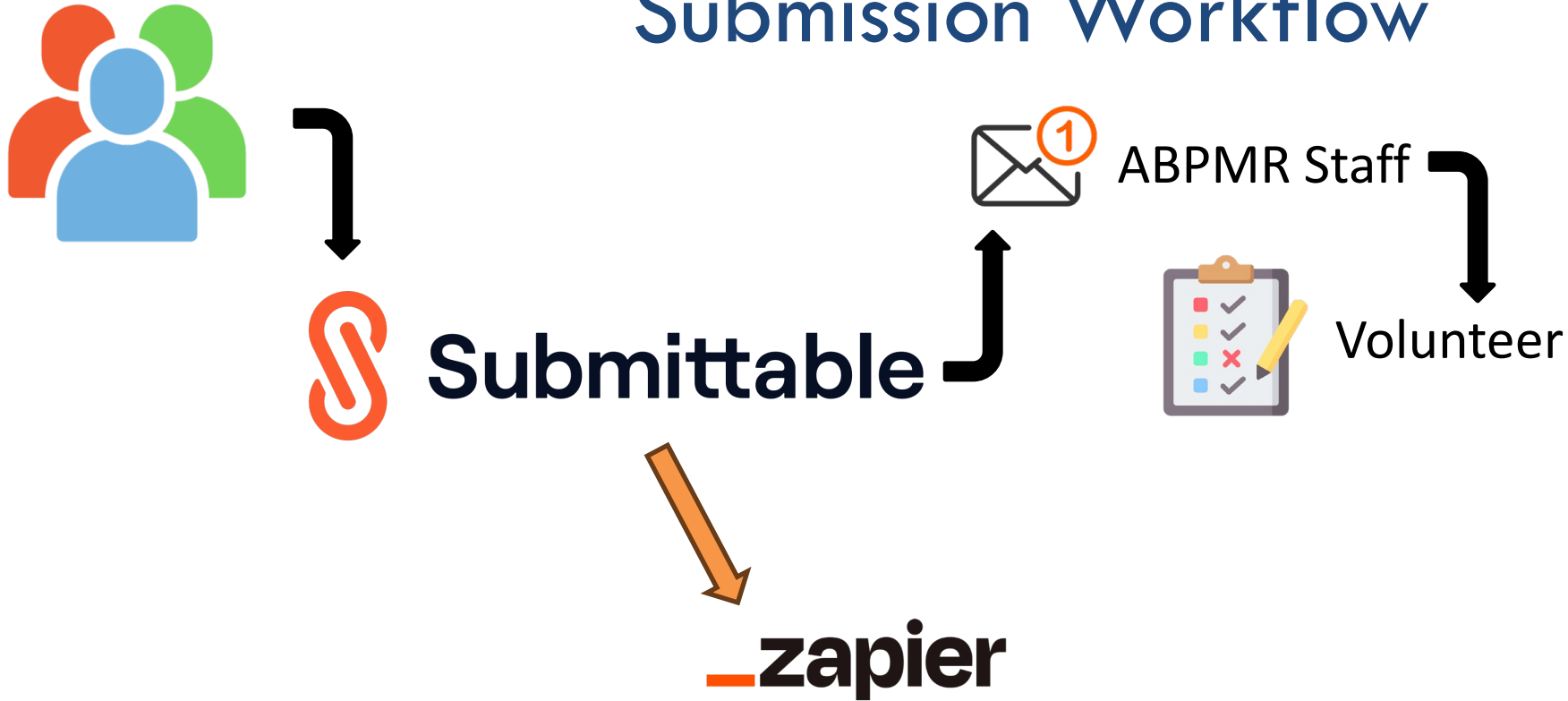
VIEWABLE BY SUBMITTER
Wed, Sep 2, 2026 1:00 PM
Satwinder Kaur sent a message

VIEWABLE BY SUBMITTER
Wed, Sep 2, 2026 1:00 PM
Satwinder Kaur changed the status to In-Progress

VIEWABLE BY SUBMITTER
Wed, Sep 2, 2026 1:00 PM
Assigned to David Pruitt by Satwinder Kaur

VIEWABLE BY SUBMITTER
Tue, Sep 1, 2026 10:19 AM
Moved to **Initial Review** automatically

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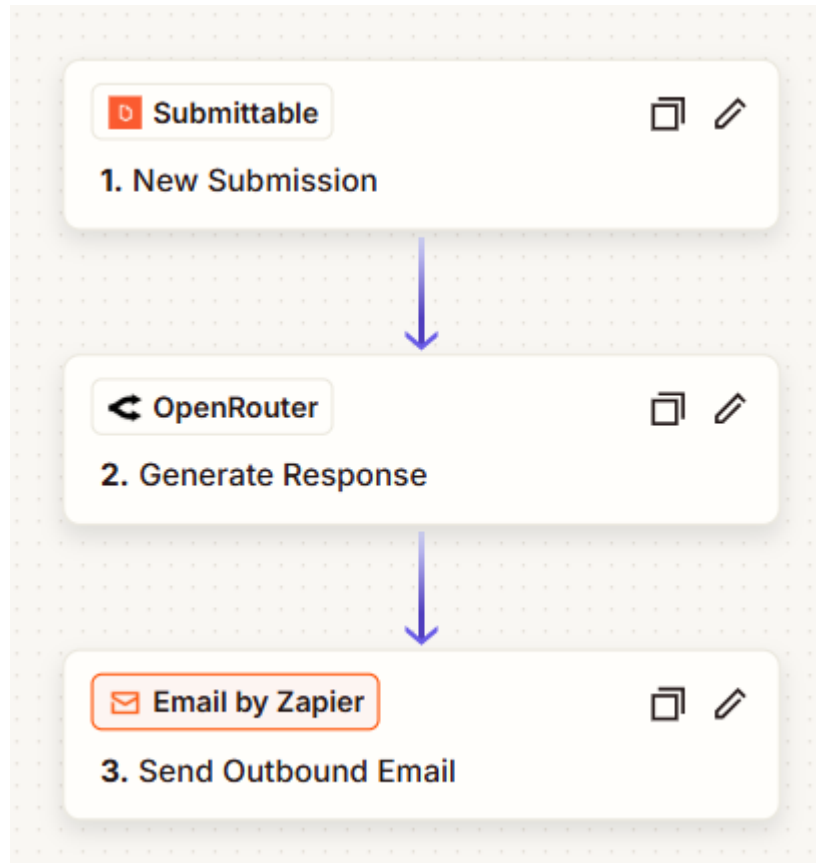
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—zapier



2. Generate Response

Next: Step 3

System Prompt

Evaluate user submissions for projects to see if these guidelines are met:
"The IHHC (QI) project should follow the well-known IHHC Plan-Do-Study-Act (PDSA) methodology that involves identifying a problem, taking a baseline measurement, setting a goal, making a change, and measuring again to see if the goal was reached."
Minimum Acceptance Criteria;
A submission should be ACCEPTED if it clearly includes all of the following:
1. Problem identified relevant to practice/patients.
2. Baseline measurement (quantitative pre-change data). Acceptable measures include outcome, process, or balancing metrics.
3. Implemented change(s) (a concrete workflow/process intervention actually carried out).
4. Follow-up measurement after the change showing results and whether the goal was met (pre/post data or equivalent).
Sometimes users do not provide data in the expected fields, but if the data is present anywhere in the submission and clearly shows a baseline and follow-up measurement, that is acceptable.
Reply with a brief determination on if the project meets such standards or if something is missing.
A human will review your determination, but your goal is to save human review's time from submissions that clearly fall short.
Be concise; no extra commentary. Do not invent data. Base decisions strictly on the document content. Do not use markup in your response.
When the submission is not accepted, or if you spot something that could improve the project, include a separate recommendation outside of your determination with modifications the submitter could make to meet the guidelines in the style of a mentor.

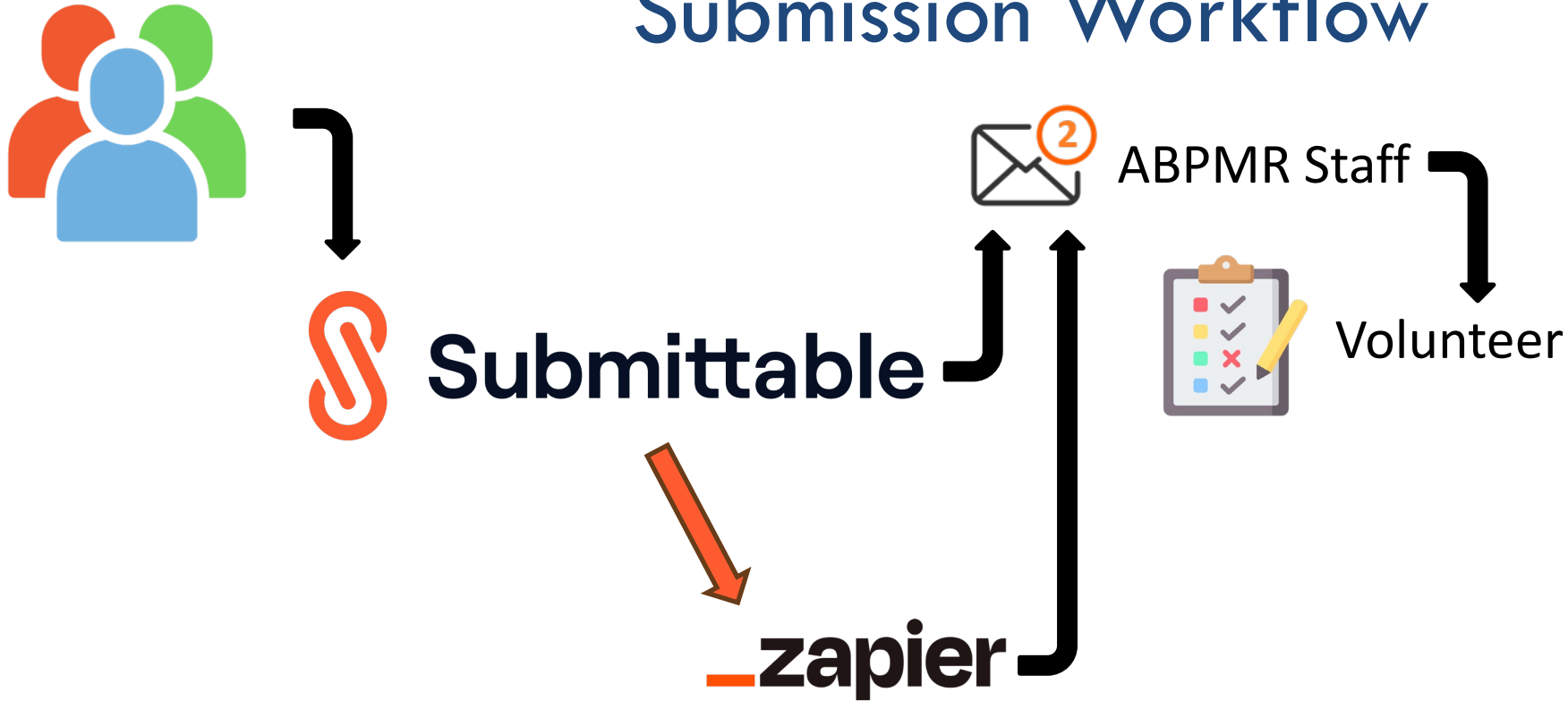
Prompt

Object.to_json(1. Raw Output: {...})

Model

Google: Gemini 3.1 Pro Preview

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Presented POC to Board in January 2026

- Discussions around process integration with volunteers
 - 2026 also introduced non-BOD volunteer reviewers
 - Decided to move into a Pilot Phase
- AI reviewer went live February 12, 2026
 - Used Gemini Pro 3.0
- Met in March with Subcommittee chair
 - High AI rejection rate due to strict SMART goal criteria

Strict SMART Goal Criteria

- **Missing Criteria:**
The submission lacks a clearly defined SMART goal (Specific, Measurable, Achievable, Relevant, Time-bound). While an "opportunity statement" is provided ("improving functional outcomes..."), it does not set a specific numerical target or a timeframe for completion.

Strict SMART Goal Criteria

- Human reviewers did not require SMART goals
 - Identified problem needs to be quantifiable
 - Baseline measurements are required
 - Follow-up measurements are required
- Adjusted the LLM guidance
 - Removed strict SMART goal requirement
 - Included guidance to suggest recommendations for rejected projects
 - Moved to Gemini Pro 3.1

Pilot Progress and AI Cost

- Live Feb 12. Guidance unchanged since March 17.
- Processed: 700+ submissions
- Total Costs
 - Zapier Pro Plan \$30/mo
 - LLM total: \$12.30
 - 2.3M tokens processed (Gemini 3.1 Pro: \$2/\$12 in/out price per 1M)
 - LLM cost per submission: ~\$0.0175
 - Including Zapier Pro costs = \$0.36 per submission
 - Overall year cost estimated at \$395 (90% of cost is Zapier)

Is it working?

- By July, 553 total submissions
- 156 processed with v1 of AI guidance
 - No detailed analysis was done for this group
- 397 processed with v2 of AI guidance
 - 268 accepted, 129 not accepted (32%)
- 528 ultimately accepted, 25 ultimately rejected
 - Human reviewers work with diplomates to address missing content

Is it working?

- 129 rejections
 - Most common: **2 missing elements** (58 projects / 45%)
 - 1 missing element: 35 projects (27%)
 - 3 missing elements: 30 projects (23%)
 - 4 or 5 missing elements: only 6 projects
- Average was **2.1 missing elements** per rejected submission.

What was missing	Number of projects	% of all rejections
Quantitative baseline	107	83%
Quantitative follow-up	92	71%
Both baseline and follow-up	80	62%
Evidence that a real change was actually implemented	42	33%
Specific metrics / sample sizes / numerator-denominator details	~20	~15%

But, IS IT WORKING?

- Goal was to reduce volunteer workload
 - Hard to quantify! If this were an IHHC/QI project, the AI reviewer would probably reject the submission
- AI Reviewer was introduced alongside non-board member volunteers
 - Fundamental shift in workflow
 - Board members now also have rookie reviewers to train and coordinate work with

Defining Success

- Board discussions are around HOW to use the AI Reviewer
 - Read AI assessment first?
 - Read diplomate submission first?
 - Copy/paste recommendations from AI reviewer?
 - Do rookie reviewers get to see the AI reviewer assessment?

It IS working!

- Board members say:
 - AI Reviewer makes the process easier and faster
 - Summary gets them ready to read the project
 - Summary confirms their understanding of the submission
 - Conversations around rookie reviewers being allowed to use it since the tool is so helpful to the board members

Why This Was a Great First Step

- Text-based interaction
- Low-risk
- Verifiable work
- Human in the loop

Project Future

- Expected to part of 2027 workflow
- Gemini 3.1 Pro will be replaced with a newer LLM
- Exploring in-house software harness to replace Zapier

Model Performance

- POC used GPT-5, first implementation used Gemini 3.0 Pro, currently using Gemini 3.1 Pro
- We do not have a way to regression test models
 - The initial POC was done by hand
 - Ideally, we'd keep a test deck for evaluating new models
- True Frontier Models are likely to be better all around
 - With average of 2k tokens in and 1k tokens out, est cost: \$0.07/submission at Fable or Astra pricing (\$10/\$50)

In-House Software Harness

- Zapier receives webhook calls from Submittable, sends prompt to OpenRouter, formats prompt into email to staff
- Zapier could easily be replaced by an in-house harness and 90% of the project cost could be reduced

Project Radar

- Exploring using LLM to process Part II Oral Exam transcripts
 - Candidate Scoring
 - Examiner Oversight
- Did a POC for an AI voice-to-voice vignette
 - Role play during oral exam
 - Professionalism assessment as part of Longitudinal Assessment

Questions?

